# Anchored price on PDP

*Adding an anchored 'compare-at' price next to the sale price on the product detail page increases conversion by making the discount more salient at the moment of purchase intent.*

AT A GLANCE

**Primary metric is *conversion_rate*; decision is binary against Ship if conversion lift CI excludes zero on benefit side.**

*Pre-registered before any analysis. Decision is binary against conversion_rate: a 5.0% relative relative lift over a 10.0% baseline requires 3,200 per arm at the standard a / power settings. Guardrails sit on revenue_per_user.*

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SAMPLE REQUIRED (PER ARM)

**3,200**

*to detect a 5.0% relative relative effect at a = 0.05, power = 0.80*

BASELINE
10.0%

TOTAL SAMPLE
6,400

DURATION
14 days

FEASIBILITY
VIABLE

MECHANISM

## Why we think this works

*Price anchoring is a well-studied behavioral effect: showing the original price increases perceived savings even when the actual discount is unchanged. This should lift conversion on items that are already on sale.*

*Sale items are ~30% of catalog impressions. Last 2025Q4 anchored-pricing test on the cart page showed flat conversion with a small AOV uplift.*

METRICS

## conversion_rate is the verdict-driving metric.

*Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.*

| | |
|---|---|
| PRIMARY | **conversion_rate**<br>Proportion · baseline 10.0% · MDE 5.0% relative · direction higher is better |
| SECONDARY | **add_to_cart_rate**<br>Leading indicator. Not used for the ship/no-ship decision but informs interpretation. |
| SECONDARY | **average_order_value**<br>Leading indicator. Not used for the ship/no-ship decision but informs interpretation. |
| GUARDRAIL | **revenue_per_user**<br>Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative. |

POWER

## Detecting smaller effects requires much more sample.

*To resolve a 5.0% relative relative effect against a 10.0% baseline, we need 3,200 per arm. Tighter detection thresholds would climb the curve to the left — quickly.*




*Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.*

ASSIGNMENT + COHORTS

## First-exposure attribution at the user_id level.

*Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.*

| | |
|---|---|
| UNIT | **user_id**<br>One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed. |

| | |
|---|---|
| ARMS | **50 / 50**<br>control · treatment |
| COHORT | **all assigned users** |

DECISION RULES

## Pre-registered. The analyze verb walks the same tree against the same metrics.

*Ship if conversion lift CI excludes zero on benefit side.*

| | |
|---|---|
| SHIP | Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7 |
| LIFT WITH CAVEAT | Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out |
| NO-SHIP · GUARDRAIL | Any guardrail's 90% CI lies past its harm threshold |
| NO-LIFT | Well-powered, primary CI straddles zero with half-width > 2 × MDE |
| INCONCLUSIVE | Underpowered (n < required) AND primary CI straddles zero |
| INVALID | SRM $\chi^2$ $p < 0.0005$ and no override resolved — investigate assignment |

INTEGRITY LOCK · THREE-PART SEAL

## brief.sealed.yaml binds intent + metric definitions + expected shape.

*Tampering with any of the three components after seal will fail `agentxp.schemas.brief_seal.verify_or_raise`; `agentxp analyze` refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.*

| | |
|---|---|
| DESIGN_CHAIN_HASH | 2b614da7fa04d573642556018fc656b6c3f03c94049365aa121e3bf74aeab646 |
| METRIC_SNAPSHOT | 1 metric hashed |
| CONVERSION_RATE | fbf7a228872bb052817eb550deb3311acba51c3cb60f03241c460211b307f2d0 |
| EXPECTED_SHAPE | user_id · 50 / 50 |
| SEALED_AT | 2026-06-05T15:43:04Z |
| SEALED_BY | walk_experiment_demo |